Xinlong Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yueze Wang, Zhen Li, Qiying Yu, et al. Emu3: Next-token prediction is all you need. *arXiv preprint arXiv:2409.18869*, 2024.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Zeqiu Wu, Yushi Hu, Weijia Shi, Nouha Dziri, Alane Suhr, Prithviraj Ammanabrolu, Noah A Smith, Mari Ostendorf, and Hannaneh Hajishirzi. Fine-grained human feedback gives better rewards for language model training. *arXiv preprint arXiv:2306.01693*, 2023.

Kevin Yang and Dan Klein. Fudge: Controlled text generation with future discriminators. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 3511–3535, 2021.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. Star: Bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems*, 35:15476–15488, 2022.

Eric Zelikman, Georges Raif Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah Goodman. Quiet-STar: Language models can teach themselves to think before speaking. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=oRXPiSOGH9.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 2019.

Biao Zhang and Rico Sennrich. Root mean square layer normalization. *Advances in Neural Information Processing Systems*, 32, 2019.

Yizhe Zhang, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Josh Susskind, and Navdeep Jaitly. Planner: Generating diversified paragraph via latent language diffusion model. *arXiv preprint arXiv:2306.02531*, 2023.

Kaiwen Zheng, Yongxin Chen, Hanzi Mao, Ming-Yu Liu, Jun Zhu, and Qinsheng Zhang. Masked diffusion models are secretly time-agnostic masked models and exploit inaccurate categorical sampling. *arXiv preprint arXiv:2409.02908*, 2024.

Chunting Zhou, Lili Yu, Arun Babu, Kushal Tirumala, Michihiro Yasunaga, Leonid Shamis, Jacob Kahn, Xuezhe Ma, Luke Zettlemoyer, and Omer Levy. Transfusion: Predict the next token and diffuse images with one multi-modal model. *arXiv preprint arXiv:2408.11039*, 2024.

Daniel M Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. Fine-tuning language models from human preferences. *arXiv preprint arXiv:1909.08593*, 2019.

## A Additional Visualizations of the Impact of Diffusion Process on Language Modeling

We provide a number of additional results in this section to help build intuition for the contribution of the diffusion model to language generation. Both Figure 2 and Figure 6, show the change in log likelihood at a token level when conditioned on a semantic plan. Figure 2 shows this for arbitrary text continuations which we embed as semantic plans with Sentence T5, while Figure 6 shows this for latent plans sampled from STAR-LDM.

In order to quantitatively assess the impact of the semantic plans, we measure the perplexity of STAR-LDM as a function of the log(SNR) in Figure 7. We do this on a subset of the C4 validation set, as well as on a set of generations nucleus sampled from STAR-LDM. We see that at high log(SNR), where the semantic plan is very clean, the perplexity is much lower.

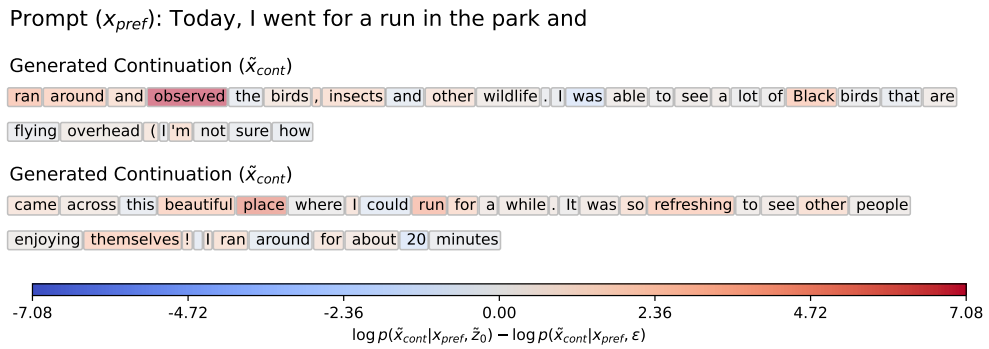


Figure 6: Visualization of the change in log likelihood due to conditioning on a semantic plan. Shown for two different generations. Positive values (red) indicate tokens that become more likely when conditioned on the plan, while negative values (blue) indicate tokens that become less likely.

However at low log(SNR), where the semantic plan is very noisy, the perplexity is much higher. This demonstrates that the semantic plan reduces the uncertainty in the language generation; helping the language model focus on a specific semantic direction.

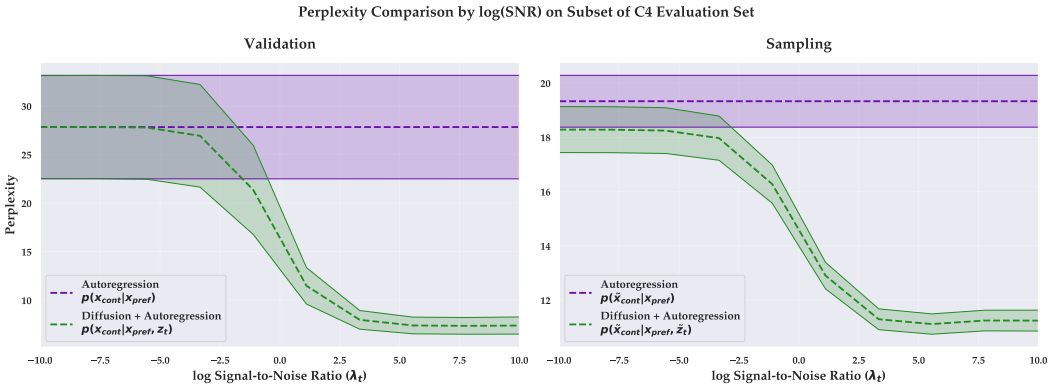


Figure 7: Perplexity at various noise levels on a subset of 100 examples from the C4 evaluation set (left), or 100 sampled generations (right).

Additionally we show the same perplexity comparison but at the maximum log(SNR), corresponding to $t = 0$, in Figure 8.

Finally to complement our quantitative evaluation of STAR-LDM across noise levels, we visualize how the token wise log likelihoods change with the log(SNR) in Figure 9.

## B Inference Latency

To explore the impact of the diffusion planning phase on efficiency, we analyze the inference latency of STAR-LDM. As the source of computational overhead is the iterative "thinking" phase, we evaluate how generation quality varies with the number of diffusion sampling steps. For this analysis, we adopt the DPM-Solver++(2M) SDE sampler (Lu et al., 2022a), which is highly effective for few-step sampling, and we sweep the number of diffusion sampling steps from 5 to 50. We set the prefix guidance to $w = 2.0$ for this study.

We generate 96-token continuations for 32-token prompts, using a set of 250 examples randomly drawn from the C4 validation set. We report generative perplexity as a function of the wall-clock time per generation in Figure 10. For reference, we also benchmark the

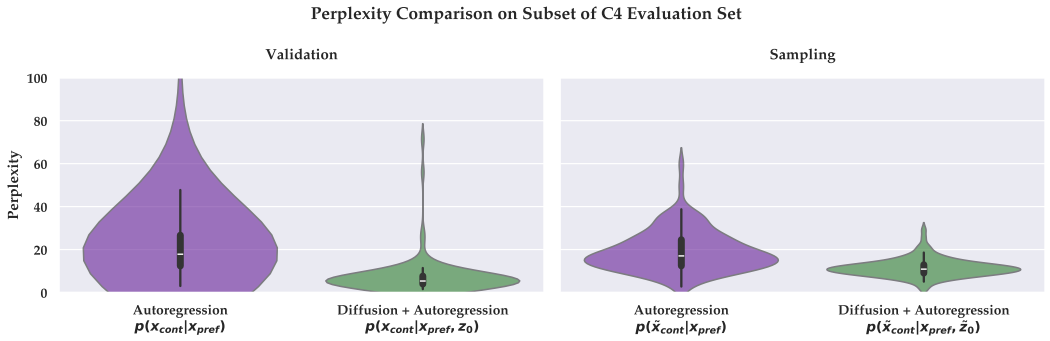


Figure 8: Perplexity on a subset of 100 examples from the C4 evaluation set (left), or 100 sampled generations (right).

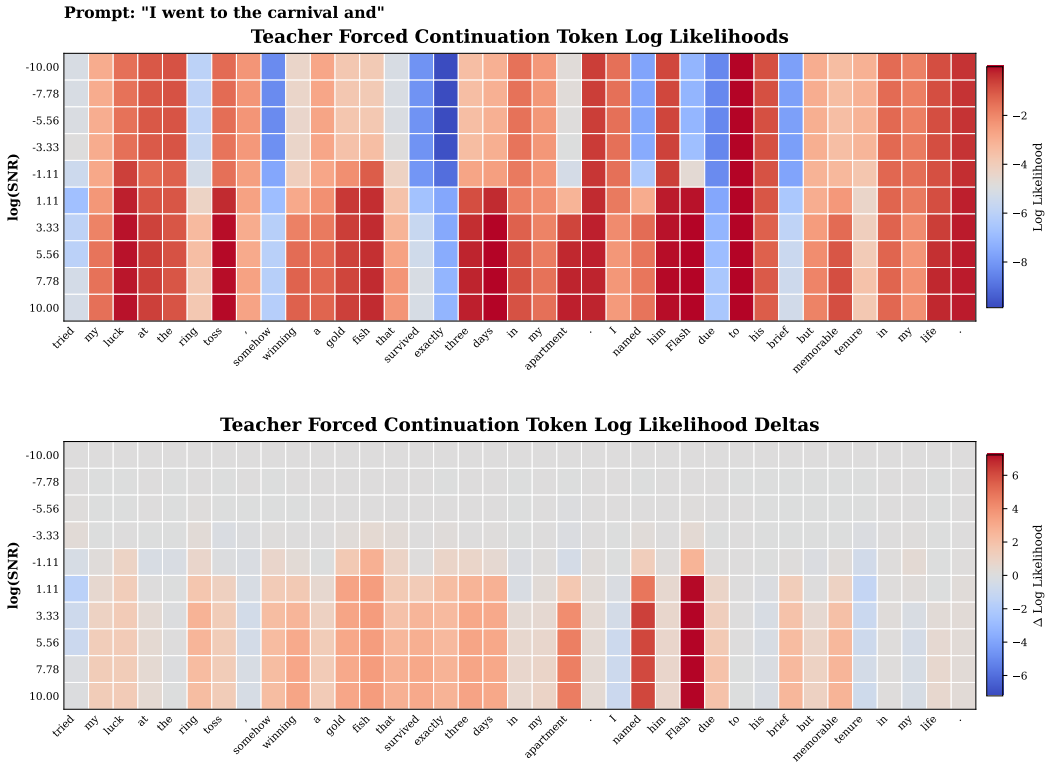


Figure 9: Visualization of the token wise log likelihood by log(SNR) (top). Visualization of the token wise change in log likelihood by log(SNR) (bottom).

inference speed and quality of the standard GPT-2 Large and GPT-2 XL autoregressive models.

It is important to note that our current STAR-LDM implementation is not optimized for inference speed. Specifically, key-value (KV) caching, a standard technique for accelerating autoregressive generation, is not yet implemented for the diffusion planning phase, during which the prefix is re-processed at each denoising step. Our reported wall-clock times for STAR-LDM are therefore a loose upper bound on its inference speed.

As shown in Figure 10, STAR-LDM introduces a tunable trade-off between computation time and generative perplexity. We observe diminishing returns as the number of sampling steps increases; perplexity improves up to around 15-20 steps and then largely plateaus,